Isotopic compositions and mineral residency of nitrogen in altered terrestrial glassy basaltic rocks: implications for astrobiology

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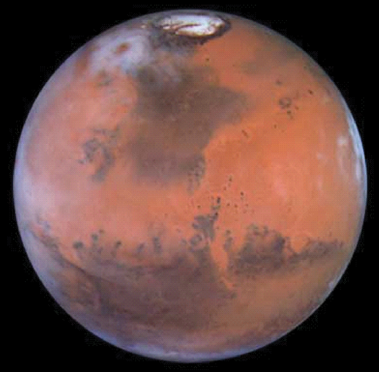
Introduction:

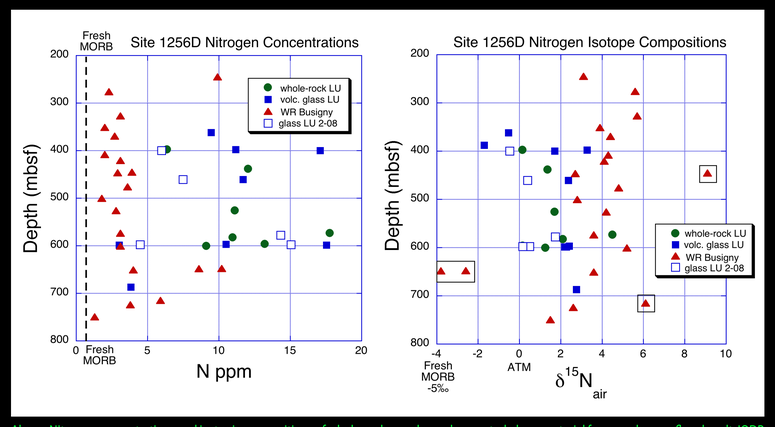
Palagonitized glasses and whole-rock samples of volcanic rocks on the modern seafloor, at ODP Site 1256D, have N concentrations of 3 to 18 ppm and $\delta^{15}N_{air}$ values of up to +5‰. Variably altered glasses from Mesozoic ophiolites (Troodos, Cyprus; Stonyford Volcanics, California, USA) contain 2 to 53 ppm N with $\delta^{15}N$ of -7.5 to +7‰. All altered glasses have N concentrations higher than those of fresh volcanic glass (for MORB glass, typically < 2 ppm N), reflecting significant N enrichment in these samples, and most of the altered glasses have $\delta^{15}N$ considerably higher than their unaltered glass equivalents (fresh MORB glass with values of -5 ± 2‰; [1]). Circulation of hydrothermal fluids, perhaps in part induced by heat energy of nearby spreading-center magmatism, could have leached NH_4^+ from nearby sediments then fixed this NH_4^+ into volcanic glasses undergoing palagonitization. Glasses from each of the three sites contain putative textural evidence for microbial activity in the form of microtubules (ichnofossils; 2-5, 11); however, any role of microbial activity in producing the N enrichments and elevated $\delta^{15}N$ values remains uncertain. A combination of petrographic analysis and imaging and chemical analyses by SEM and STEM indicates the presence of phyllosilicates (smectite and illite) in both the palagonitized cracks and the microtubules possibly of microbial origin. These phyllosilicates are the likely hosts for N in the glasses, presumably as NH_4^+. Enrichments of N (and the $\delta^{15}N$ of this N) in glasses in samples representing Earth's modern and ancient seafloor have direct relevance in considerations of modern global N subduction fluxes and ancient life. Similarly altered glasses on Mars and other extraterrestrial bodies could serve as useful biogeochemical tracers.



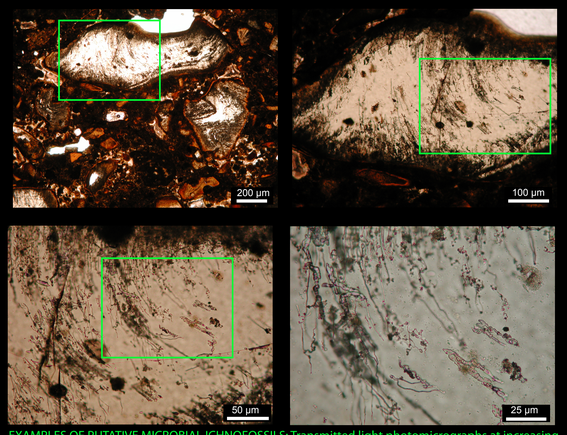
References: [1] P. Cartigny & B. Marty (2013) ELEMENTS 9(5), 359-366. [2] H. Furnes et al. (2001) Jour. Geol. Soc. 158, 75-84. [3] E. Knowles et al. (2013) EPSL 374, 239–250 [4] D. Fliegel et al. (2012) GCA 96, 80-93 [5] D. Wacey et al. (2014) Chem. Geol. 363, 1-12. [6] V. Busigny et al. (2005), G3 6 (12). [7] L. Li et al. (2007), GCA 71, 2344-2360. [8] G. E. Bebout et al. (2007) Chem. Geol. 240, 1-10. [9] N. R. Banerjee & K. Muehlenbachs, (2003) G3 4 (4). [10] N. R. Banerjee and M. R. M. Izawa, unpublished results. [11] C. M. Santelli et al. (2010) Geomicrobiol. Jour. 27, 2, 158-169. [12] N. McLoughlin et al. (2011) Chem. Geol. 289, 154-162. [13] T. Torsvik et al, (1998) EPSL 162, 165-176. [14] L. D. Anderson et al. (2010) LPS XL Abs. #1668. [15] M. Fogel & A. Steele (2013) ELEMENTS 9(5), 367-372. [16] Aoudjehane et al. (2012) Science 338, 785-788. [17] L. M. White et al. (2014) Astrobiology 14, 170-181. [18] C. Thomazo & D. Papineau (2013) ELEMENTS 9(5), 345-352.

Modern Seafloor:

Nitrogen isotope work on modern oceanic crust suggests that signatures of the addition of sedimentary/organic N are in some cases preserved. Basaltic whole-rocks and separates of variably palagonitized volcanic glass, from the modern seafloor, are enriched in N (whole-rocks, glasses up to 18.2 ppm) and have elevated $\delta^{15}N_{air}$ (up to +8.3‰) relative to that of fresh MORB (near -5‰; [also see 6,7]). N contents and $\delta^{15}N$ are consistent with addition of sedimentary/organic N via interaction with pore fluids, with or without direct mediation by biological activity. However, recent work identifying endogenous organic matter associated with microbial alteration structures possibly provides a link to microbial activity.

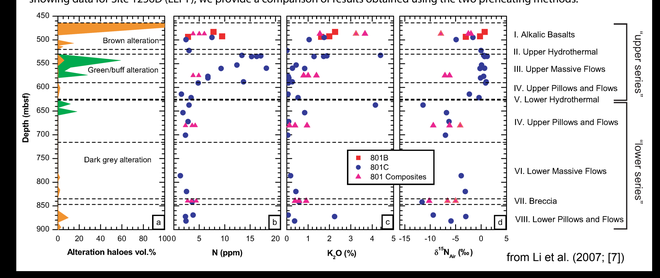


Above: Nitrogen concentrations and isotopic compositions of whole-rock samples and separated glassy material from modern seafloor basalt, IODP site1256D. Note the large shifts in both concentration and $\delta^{15}N$ relative to fresh MORB (< 2 ppm, -5‰, respectively). Also included are data from Busigny et al. [6], for comparison.



EXAMPLES OF PUTATIVE MICROBIAL ICHNOFOSSILS: Transmitted light photomicrographs at increasing magnifications showing examples of tubular bioalteration textures in basaltic glass, Ontong Java Plateau, IODP leg 192 [9]. Tubules originate on grain boundaries and are mostly hollow, with a variety of morphologies.

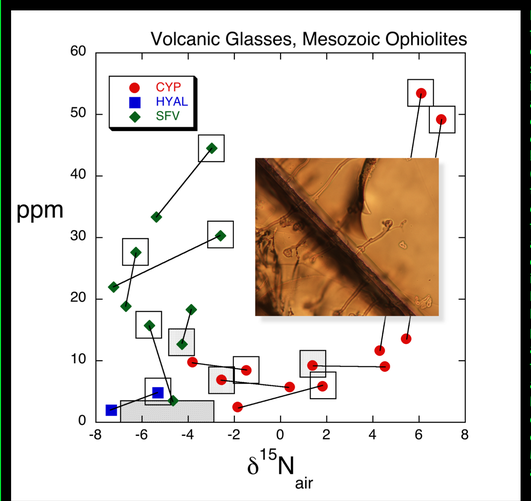
Analytical Methods: Glass N concents and isotopic compositions were analyzed using the carrier gas methods described by Bebout et al. (2007; [8]). Nitrogen extracted (in the form of N_2) was purified in an all-metal extraction line, then transferred into a Finnigan MAT 252 mass spectrometer via a Finnigan Gas Bench II and a U-trap interface where samples of N_2 were entrained in a He stream. Under a binocular microscope, glass samples were separated into clean (black and glassy) and altered (glass with dull brown encrustations) fractions and then crushed. About 100-150 mg of the clean or altered glass samples were loaded into quartz tubes with 1 g of CuO_2 reagent and evacuated for 24 hours on a vacuum manifold before sealing. Tubes were heated at ≥1050˚C for 180 minutes in a programmable furnace with the cooling history carefully regulated to ensure speciation of N as N_2. One issue involving sample treatment with particular relevance to our results in the method by which the samples were pre-heated, compared with methods used by Busigny et al. (2005). In the study by these latter authors, samples were pre-heated, prior to extractions, at a temperature of 450 ˚C in an oxidizing environment whereas, in our work on similar samples (see Bebout et al., 2007; Li et al., 2007 [7,8]), a pre-heating temperature of ~100 ˚C was selected (at high vacuum, not in an oxidizing environment). We believe that, at the preheating conditions employed by Busigny et al. (2005; [6]), a significant amount of N could be released, for example, from minerals such as clays and zeolites, both identified in altered oceanic crustal rocks. In the figures in this poster showing data for Site 1256D (LEFT), we provide a comparison of results obtained using the two preheating methods.



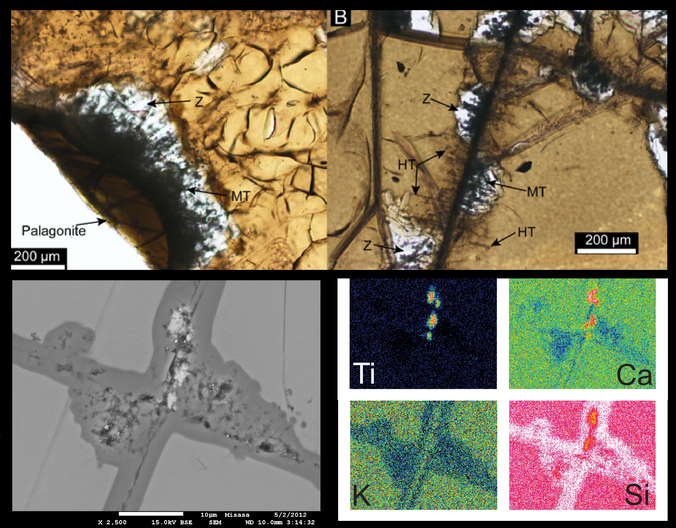
from Li et al. (2007; [7])

Mesozoic Ophiolites:

Palagonitized glasses from the Troodos ophiolite and Stonyford volcanics (SFV) typically have higher N concentration (up to 53 ppm) and $\delta^{15}N$ values than less-altered glass separates. Some SVF glasses (with and without alteration) are enriched in N relative to fresh MORB glass, but have mantle-like $\delta^{15}N$ of -8 to -4‰. We suggest that, in these samples, early-formed palagonite incorporated mantle-derived N and later-formed palagonite incorporated sedimentary/organic N introduced by pore fluids.



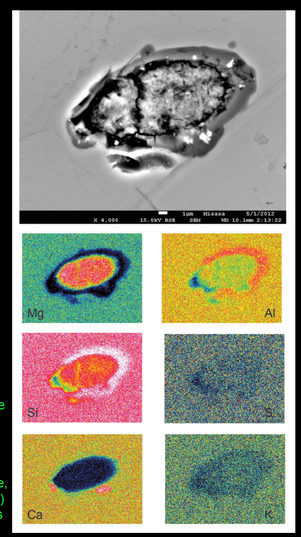
LEFT: Nitrogen concentrations and isotopic compositions of glass separates from basalts in the Troodos ophiolite and the Stonyford Volcanic Complex. Tie-lines connect altered (in boxes) with less altered (no boxes) glasses for the same sample. For 9 of the 12 pairs, the glass that appeared more altered (i.e., larger abundance light-brown coating) is more enriched in N and has higher $\delta^{15}N$. Inset photograph (200 µm wide) shows an example of a microbial tubule emanating from a palagonitized fracture. Note the range of concentrations and isotopic compositions for "fresh" MORB (grey-shaded box at bottom of figure; [1]).

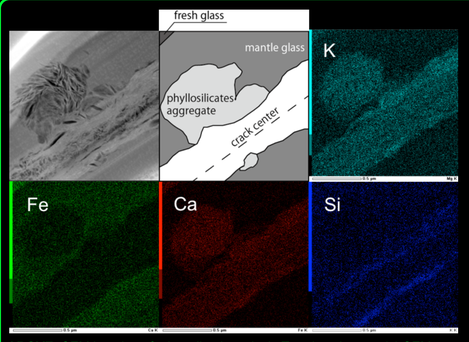


LEFT: Photomicrographs (viewed in transmitted, plane-polarized light; [10]) of alteration assemblages from the Stonyford Volcanic Complex, including multiple possible environments for nitrogen incorporation. **HT: Hollow tubules** (or mostly hollow, some contain small amounts of phyllosilicates and other alteration materials, and possibly organic compounds) within unaltered basaltic glass (yellow-brown). **MT: Mineralized tubules** (mineralized by fine-grained titanite) within zeolite-dominated alteration zones. **Z: Zeolite-dominated alteration**, Ca-phillipsite is the dominant zeolite species in these examples (others include analcime, heulandite and Ca-stilbite). In these examples, basaltic glass is not expected to be a major carrier of nitrogen due to the generally low solubility of N in basaltic liquids. Nitrogen, as NH_4^+, is likely to be incorporated into the palagonite alteration materials via substitutions of NH_4^+ for K^+, particularly in the phyllosilicate components. Ammonium substitution for alkali ions may also occur in the zeolitic alteration assemblages along with incorporation of nitrogen species as free molecules (e.g., N_2, NH_3) in the large cage sites of zeolite minerals. There may be a small contribution from nitrogen incorporated in organic compounds (see Wacey et al., 2013 [5]).

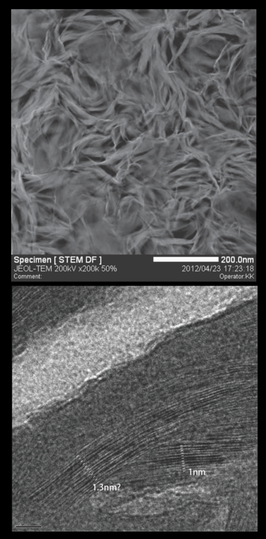
LEFT BOTTOM: SEM imaging of mineralized cracks in glasses from the Stonyford Volcanics. (a) SEM image, showing two intersecting cracks with obvious alteration. (b-i) element maps for the same region, showing element distributions in and adjacent to the cracks. The Ca-Ti-rich zones likely correspond to concentrations of titanite.

RIGHT: SEM imaging of a cross-section of an individual microtubule in a glass from the Stonyford Volcanics. (TOP) SEM image, showing two intersecting cracks with obvious alteration. (OTHER) element maps for the same region, showing element distributions in and adjacent to the microtubule.





ABOVE: SEM imaging of mineralized cracks in Troodos glasses. (a) SEM image, showing a crack in the glass with a nearby and seemingly genetically related phyllosilicate aggregate. LOWER: element maps for the same region, showing element distributions in and adjacent to the crack and in the phyllosilicate aggregate. Note the enrichments in K in the crack and aggregate, Si enrichments at/near the edge of the crack and its reaction rim, and the depletion in Fe in both the mineralized crack and the mineral aggregate. **RIGHT:** STEM images of a phyllosilicate-rich region in a Troodos volcanic glass. The lower, higher-magnification image illustrates d-spacing indicative of the presence of smectite and illite, each phase (but particularly illite) capable of incorporating N as NH_4^+. For Troodos glasses, Wacey et al. [5] confirmed the linings of some microbial tubules with C (± N), providing another possible N contribution to the bulk-glass analyses [cf. 12, 13].



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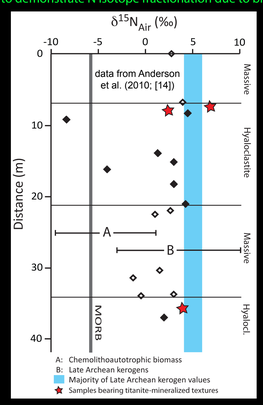
Archæan Basalts:

Abitibi 2.7 Ga proto-glassy lavas altered to greenschist-facies exhibit similar increased N contents and $\delta^{15}N$ relative to MORB. Whole-rock analyses of samples with and without ichnofossils yield mostly positive $\delta^{15}N$ up to +6.9‰. $\delta^{15}N$ of whole-rock samples from massive and proto-glassy flow units is mostly in the range for Late Archean kerogens (-6 to +12‰), which represent organic N that escaped mineralization in marine sediments [14]. Again, the enrichment of ^{15}N in the Archean basaltic rocks demonstrates alteration by N-bearing fluids carrying sedimentary/organic $\delta^{15}N$ signatures. The putative ichnofossil-bearing samples appear not to host unambiguous N-isotope signatures representative of microbial processing, not surprising as little or no biomass remains in the tubular structures. However, whole-rock $\delta^{15}N$ of the Archean basalts appears to demonstrate N isotope fractionation due to biologic denitrification.



LEFT: $\delta^{15}N$ of whole-rock metabasalts in an Abitibi Complex exposure at the Hurd Locality, near Rouyn, Quebec (studied by L. D. Anderson, Lehigh M.S. student [14]). This nearly 100% exposure allows the sampling of low-grade 2.7 Ga basalts across well-characterized flow units, in some of which Si-rich hyaloclastite zones contain putative microbial ichnofossils (see red stars, [14]). Shown on this plot are the $\delta^{15}N$ of fresh MORB and the ranges in $\delta^{15}N$ of chemolithoautotrophic biomass and Late Archean kerogens (see [18]).

Conclusions:

The results presented here constitute an exploratory study of the incorporation of N, through low-T hydrothermal alteration, into volcanic glass from the modern seafloor and Mesozoic volcanic glasses from Cyprus and the Franciscan ophiolites. Palagonitized ophiolite glasses have higher N concentrations (up to 53 ppm) and $\delta^{15}N$ values than less-altered glass separates from the same samples. All altered glasses have N concentrations higher than those of fresh MORB (typically less than 2 ppm N), reflecting significant N enrichment in these glass samples relative to the concentrations that can be attributed to magmatic processes. The observed shifts in N contents and $\delta^{15}N$ can be attributed to incorporation of sedimentary/organic N via pore fluids. Circulation of hydrothermal fluids, perhaps induced by heat from the nearby magmatism, could have leached NH_4^+ from coexisting and overlying sediments then fixed to glasses. This alteration could continue during the longer-term transit of oceanic crust across ocean basins and toward subduction zones at which it contributes to the subducting seafloor N inventory. Although the samples investigated in this study in many cases contain textural evidence for microbial alteration, and considerable mineralization occurs in these microtubules, direct association of the N enrichments with this microbial alteration awaits a microanalytical method for analyzing the isotopic compositions of the N at the scale investigated in recent microanalytical imaging and spectroscopic studies (e.g., Wacey et al., 2013 [5]).

These findings should be taken into consideration in planning strategies for the search for (modern and ancient) life on Mars and other extraterrestrial bodies (see the broader discussion by Fogel and Steele, 2013 [15]). The enhanced chemical reactivity of volcanic glass makes it a potentially useful receptacle for biologically (or non-biologically) processed N that could yield information regarding biogeochemical pathways at the time of incorporation. Future Mars sample return missions should prioritize return of palagonitized basaltic rocks with the hope that the glassy parts of these rocks retain information regarding modern or ancient life. Investigation of the silicate N isotope composition of martian meteorites that have experienced aqueous alteration (e.g., Tissint meteorite [15,16]; Yamato meteorite (White et al., 2014; [17]) can potentially provide insights regarding martian surface N cycles, including their temporal evolution.